Advanced Plane Geometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Circle theorems: tangent-chord and secant angles**

Two secants are drawn from an external point to a circle, intercepting arcs of 100° and 30°. Find the measure of the angle formed at the external point, using the formula (difference of arcs) ÷ 2.

Angle $= \frac{100^\circ - 30^\circ}{2} = \frac{70^\circ}{2} = 35^\circ$.

2 **[Foundational]** • **Target: Circle theorems: tangent-chord and secant angles**

Two chords intersect inside a circle, intercepting arcs of 80° and 40°. Find the measure of the angle formed at the intersection, using the formula (sum of arcs) ÷ 2.

Angle $= \frac{80^\circ + 40^\circ}{2} = \frac{120^\circ}{2} = 60^\circ$.

3 **[Foundational]** • **Target: Power of a point (secant and tangent segment relationships)**

A tangent segment from an external point has length 8, and a secant from the same point passes through the circle with external segment length 4. Find the length of the entire secant segment, using $(\text{tangent})^2 = (\text{external secant})(\text{whole secant})$.

$8^2 = 4 \times (\text{whole secant})$, so $64 = 4w$, giving $w = 16$.

4 **[Foundational]** • **Target: Area of regular polygons**

Find the area of a regular pentagon with side length 6 and apothem 4.13, using $A = \frac{1}{2} \times \text{perimeter} \times \text{apothem}$.

Perimeter $= 5(6) = 30$. Area $= \frac{1}{2}(30)(4.13) = 61.95$.

5 **[Foundational]** • **Target: Similar solids and scale factor in 3D**

Two similar cylinders have a scale factor of $\frac{2}{3}$. If the smaller cylinder has volume 40, find the volume of the larger cylinder.

Volumes of similar solids scale by the CUBE of the linear scale factor: $\left(\frac{3}{2}\right)^3 = \frac{27}{8}$. Larger volume: $40 \times \frac{27}{8} = 135$.

6 **[Foundational]** • **Target: Composite 3D solids**

A solid consists of a cylinder of radius 3 and height 8, topped with a hemisphere of the same radius. Find the total volume, in terms of π.

Cylinder volume: $\pi r^2 h = \pi(9)(8) = 72\pi$. Hemisphere volume: $\frac{2}{3}\pi r^3 = \frac{2}{3}\pi(27) = 18\pi$. Total volume: $72\pi + 18\pi = 90\pi$.

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7 [Foundational] • Target: Triangle centers: centroid and circumcenter

Find the centroid of the triangle with vertices $(2, 3)$, $(8, 3)$, $(5, 9)$, using the formula (average of the vertex coordinates).

Centroid $= \left(\frac{2+8+5}{3}, \frac{3+3+9}{3}\right) = \left(\frac{15}{3}, \frac{15}{3}\right) = (5, 5)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Heron's formula

A triangle has sides 7, 8, and 9. Find its area using Heron's formula, $A = \sqrt{s(s-a)(s-b)(s-c)}$ where s is the semi-perimeter.

$s = \frac{7+8+9}{2} = 12$. Area $= \sqrt{12(12-7)(12-8)(12-9)} = \sqrt{12(5)(4)(3)} = \sqrt{720} = 12\sqrt{5}$.

9 [Intermediate] • Target: A multi-part circle geometry problem

This question has two parts. A circle has radius 10. A chord is drawn 6 units from the center. (a) Find the length of the chord, using the relationship between the radius, the distance from center to chord, and half the chord length (a right triangle relationship). (b) Find the area of the minor segment cut off by this chord, to the nearest whole number, given that the central angle subtended by the chord is approximately 106.3° and $\sin(106.3^\circ) \approx 0.960$.

(a) Half the chord length: $\sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8$. Full chord length: 16. (b) Segment area = sector area – triangle area. Sector area: $\frac{106.3}{360} \times \pi(10)^2 \approx 0.2953 \times 314.16 \approx 92.8$. Triangle area: $\frac{1}{2}(10)(10)\sin(106.3^\circ) \approx 50(0.960) = 48$. Segment area: $92.8 - 48 \approx 45$.

10 [Intermediate] • Target: Angle bisectors and the angle bisector theorem

In triangle ABC, the angle bisector from A meets side BC at point D, dividing it such that $BD = 6$ and $DC = 9$. If $AB = 8$, find AC, using the angle bisector theorem ($\frac{BD}{DC} = \frac{AB}{AC}$).

$\frac{6}{9} = \frac{8}{AC}$, so $6 \cdot AC = 72$, giving $AC = 12$.

11 [Intermediate] • Target: The 30-60-90 and 45-45-90 triangles in composite figures

An equilateral triangle has side length 10. Find its height and area, using the fact that the height splits it into two 30-60-90 triangles.

The height corresponds to the longer leg of a 30-60-90 triangle with hypotenuse 10 and shorter leg 5: height $= 5\sqrt{3}$. Area $= \frac{1}{2}(10)(5\sqrt{3}) = 25\sqrt{3}$.

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12 **[Intermediate]** • **Target: Circumscribed and inscribed circles**

A square is inscribed in a circle of radius 5. Find the side length of the square, using the fact that the diagonal of the square equals the diameter of the circle.

Diameter $= 2(5) = 10$. For a square, diagonal $= s\sqrt{2}$, so $s\sqrt{2} = 10$, giving $s = \frac{10}{\sqrt{2}} = 5\sqrt{2}$.

13 **[Intermediate]** • **Target: Regular polygon interior and exterior angles**

Find the measure of each interior angle and each exterior angle of a regular octagon.

Sum of interior angles of an n-gon: $(n-2)180^\circ$. For $n = 8$: $(6)(180^\circ) = 1080^\circ$. Each interior angle: $\frac{1080^\circ}{8} = 135^\circ$. Each exterior angle (supplementary to the interior angle): $180^\circ - 135^\circ = 45^\circ$.

14 **[Intermediate]** • **Target: A multi-part composite figure and circle problem**

This question has three parts. A running track consists of a rectangle with two semicircular ends. The rectangle is 80 meters long and 40 meters wide, and the semicircles are attached to the two 40-meter-wide ends. (a) Find the radius of each semicircle. (b) Find the total perimeter (distance around) the track, in terms of π. (c) Find the total enclosed area of the track shape, in terms of π.

(a) The semicircles are attached to the 40-meter ends, so the diameter of each semicircle is 40, giving radius $= 20$. (b) Perimeter $= 2(\text{rectangle length}) + 2(\text{semicircle arc length})$. The two semicircles together form one full circle's circumference: $2\pi(20) = 40\pi$. Total perimeter: $2(80) + 40\pi = 160 + 40\pi$ meters. (c) Area $=$ rectangle area $+$ combined semicircle area. Rectangle area: $80 \times 40 = 3200$. The two semicircles together form one full circle: $\pi(20)^2 = 400\pi$. Tota.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Power of a point (secant and tangent segment relationships)** • **Cross-Lesson: 8.01 Statistics, Counting & Probability**

Two chords AB and CD intersect inside a circle at point X. If $AX = 4$ cm, $XB = 9$ cm, and $CX = 3$ cm, calculate the length of segment XD and the total length of chord CD.

Step 1 (Intersecting Chords Theorem): $AX \cdot XB = CX \cdot XD$ $4 \times 9 = 3 \times XD \Rightarrow 36 = 3 \cdot XD \Rightarrow XD = \frac{36}{3} = \mathbf{12}$ cm
Step 2 (Chord Length): $CD = CX + XD = 3 + 12 = \mathbf{15}$ cm

16 **[Challenging]** • **Target: Area of regular polygons** • **Cross-Lesson: 1.01 Advanced Number Theory**

A regular hexagon has a side length of $s = 8$ cm. Find the length of its apothem a and its exact total area A.

Step 1 (Apothem): A regular hexagon is composed of 6 equilateral triangles of side length $s = 8$. The apothem is the altitude of one equilateral triangle: $a = s\cos(30^\circ) = 8\left(\frac{\sqrt{3}}{2}\right) = \mathbf{4\sqrt{3}}$ cm ≈ 6.928 cm Step 2 (Area): Using the perimeter $P = 6 \times 8 = 48$ cm: $A = \frac{1}{2}aP = \frac{1}{2}(4\sqrt{3})(48) = 24 \times 4\sqrt{3} = \mathbf{96\sqrt{3}}$ cm² ≈ 166.28 cm²

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17 **[Challenging]** • **Target: Similar solids and scale factor in 3D** • **Cross-Lesson: 2.01 Advanced Algebra & Systems**

A right circular cone has a base radius of 6 cm and a slant height of 10 cm. Find (a) its perpendicular vertical height h, (b) its volume V in terms of π, and (c) its total surface area in terms of π.

(a) Vertical height: Using the Pythagorean relation $l^2 = r^2 + h^2$:
$h = \sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = \mathbf{8}$ cm (b) Volume:

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(6^2)(8) = \frac{1}{3}\pi(36)(8) = 12 \times 8\pi = \mathbf{96\pi}$ cm³ ≈ 301.59 cm³ (c) Total surface area:

TSA $= \pi r^2 + \pi r l = \pi(6^2) + \pi(6)(10) = 36\pi + 60\pi = \mathbf{96\pi}$ cm² ≈ 301.59 cm²

18 **[Challenging]** • **Target: Composite 3D solids** • **Cross-Lesson: 3.01 Advanced Functions**

A tangent from an external point P touches a circle at T. If the distance from P to the circle's centre O is 13 and the radius is 5, find the length of the tangent segment PT.

Step 1: Radius OT is perpendicular to tangent line PT at point of tangency T, forming right triangle ΔOPT with hypotenuse $OP = 13$ and leg $OT = 5$. Step 2: Apply the Pythagorean theorem:
$PT = \sqrt{OP^2 - OT^2} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = \mathbf{12}$

19 **[Challenging]** • **Target: Triangle centers: centroid and circumcenter** • **Cross-Lesson: 4.01 Advanced Trigonometry**

A triangle has side lengths $a = 5$, $b = 6$, and $c = 7$. Use Heron's formula to compute its exact area.

Step 1: Compute semi-perimeter s: $s = \frac{a+b+c}{2} = \frac{5+6+7}{2} = \frac{18}{2} = 9$. Step 2: Apply Heron's formula:

Area $= \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{9(9-5)(9-6)(9-7)} = \sqrt{9(4)(3)(2)} = \sqrt{216} = \mathbf{6\sqrt{6}} \approx 14.70$.

20 **[Challenging]** • **Target: Heron's formula** • **Cross-Lesson: 5.01 Pre-Calculus Concepts**

Find the sum of interior angles of a polygon with 9 sides.

Apply the polygon interior angle sum formula $S = (n-2) \times 180^\circ$ with $n = 9$:
$S = (9-2) \times 180^\circ = 7 \times 180^\circ = \mathbf{1260}^\circ$ Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Circle theorems: tangent-chord and secant angles

Find the area of a rectangle with length 19 and width 18.

a. 74

b. **342** ✓

c. 360

d. 323

Area $= l \times w = 19 \times 18 = 342$.

22 [Foundational] • Target: Power of a point (secant and tangent segment relationships)

Find the measure of each interior angle of a regular polygon with 18 sides.

a. 169

b. 163

c. 162

d. **160** ✓

Each interior angle $= \frac{(n-2) \times 180^\circ}{n} = \frac{(18-2) \times 180^\circ}{18} = 160^\circ$.

23 [Foundational] • Target: Area of regular polygons

Two similar solids have a linear scale factor of $2:1$. If the surface area of the smaller solid is 12, find the surface area of the larger solid.

a. 14

b. 2

c. 13

d. **3** ✓

Surface-area ratio $= \left(\frac{1}{2}\right)^2 \Rightarrow SA_2 = 12 \times \frac{1}{4} = 3$.

24 [Foundational] • Target: Similar solids and scale factor in 3D

Find the measure of each exterior angle of a regular polygon with 12 sides.

a. 36

b. **30** ✓

c. 32

d. 21

Each exterior angle $= \frac{360^\circ}{n} = \frac{360^\circ}{12} = 30^\circ$.

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25 [Foundational] • Target: Composite 3D solids

A regular polygon has an exterior angle of 36°. How many sides does it have?

a. 6

b. **10** ✓

c. 7

d. 4

$n = \frac{360^\circ}{36^\circ} = 10$.

26 [Foundational] • Target: Triangle centers: centroid and circumcenter

Find the perimeter of a rectangle with length 9 and width 20.

a. 180

b. 60

c. 29

d. **58** ✓

Perimeter $= 2(l + w) = 2(9 + 20) = 58$.

27 [Foundational] • Target: Heron's formula

In triangle ABC, point D lies on side BC such that AD is an angle bisector of angle A. Let $AB = c$, $AC = b$, $BD = m$, and $CD = n$. Prove that $\frac{m}{n} = \frac{c}{b}$ using a combination of the Sine Rule and the properties of angle bisectors.

a. **This statement is a direct consequence of the Sine Rule applied to triangles ABD and ACD.** ✓

b. This statement can be proven by applying the Law of Cosines to triangles ABD and ACD and equating expressions for AD^2.

c. This statement is true only if triangle ABC is isosceles with $b = c$.

d. This statement requires the use of vector methods for its proof.

Let $\angle BAD = \angle CAD = \alpha$. In triangle ABD, by the Sine Rule, we have $\frac{m}{\sin\alpha} = \frac{c}{\sin\angle ADB}$. Therefore, $m = \frac{c\sin\alpha}{\sin\angle ADB}$. In triangle ACD, by the Sine Rule, we have $\frac{n}{\sin\alpha} = \frac{b}{\sin\angle ADC}$. Therefore, $n = \frac{b\sin\alpha}{\sin\angle ADC}$. Since $\angle ADB$ and $\angle ADC$ are supplementary angles, $\sin\angle ADB = \sin(180^\circ - \angle ADC) = \sin\angle ADC$. Now, we can form the ratio.

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28 [Foundational] • Target: A multi-part circle geometry problem

In the diagram below, two similar triangles ABC and ADE are shown, with point D on segment AB and point E on segment AC. If AD = 4, DB = 8, and AE = 6, what is the length of EC?

a. **12** ✓

b. 10

c. 8

d. 9

The problem states that triangle ABC is similar to triangle ADE. This means that the ratios of corresponding sides are equal. We are given AD = 4, DB = 8, so AB = AD + DB = 4 + 8 = 12. We are also given AE = 6. We need to find the length of EC. Since the triangles are similar, we have the proportion: $\frac{AD}{AB} = \frac{AE}{AC}$ Substituting the given values: $\frac{4}{12} = \frac{6}{AC}$ Simplify the left side: $\frac{1}{3} = \frac{6}{AC}$ Cross-multiply to solve for AC: $AC = 3 \times 6 = 18$ We know that AC =.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Angle bisectors and the angle bisector theorem

In the diagram below, triangle ABC is a right-angled triangle with the right angle at C. An altitude is drawn from C to the hypotenuse AB, meeting AB at point D. If AC = 9 and BC = 12, find the length of the altitude CD.

a. **7.2** ✓

b. 10.8

c. 8.4

d. 9.6

Let CD be the altitude from C to the hypotenuse AB in the right-angled triangle ABC. We are given AC = 9 and BC = 12. First, we find the length of the hypotenuse AB using the Pythagorean theorem: $AB^2 = AC^2 + BC^2 = 9^2 + 12^2 = 81 + 144 = 225$. Thus, $AB = \sqrt{225} = 15$. The area of triangle ABC can be calculated in two ways: $\frac{1}{2} \times AC \times BC$ and $\frac{1}{2} \times AB \times CD$. Equating these two expressions for the area, we get.

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30 [Intermediate] • Target: The 30-60-90 and 45-45-90 triangles in composite figures

The three angles of a triangle measure $(3x - 5)^\circ$, $(3x - 10)^\circ$, and $(x + 20)^\circ$. Find the measure of the largest angle.

a. 70 ✓

b. 64

c. 62

d. 65

$(3x - 5) + (3x - 10) + (x + 20) = 180 \Rightarrow x = 25$. Angles: 70°, 65°, 45°; largest $= 70^\circ$.

31 [Intermediate] • Target: Circumscribed and inscribed circles

Two similar solids have a linear scale factor of $1:4$. If the volume of the smaller solid is 13, find the volume of the larger solid.

a. 443

b. 655

c. 832 ✓

d. 193

Volume ratio $= \left(\frac{4}{1}\right)^3 \Rightarrow V_2 = 13 \times \frac{64}{1} = 832$.

32 [Intermediate] • Target: Regular polygon interior and exterior angles

What is the sum of the interior angles of a convex polygon with 4 sides?

a. 367

b. 360 ✓

c. 368

d. 358

Sum $= (n - 2) \times 180^\circ = (4 - 2) \times 180^\circ = 360^\circ$.

33 [Intermediate] • Target: A multi-part composite figure and circle problem

The sum of the interior angles of a convex polygon is 2520°. How many sides does the polygon have?

a. 20

b. 16 ✓

c. 19

d. 3

$(n - 2) \times 180 = 2520 \Rightarrow n - 2 = 14 \Rightarrow n = 16$.

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34 [Intermediate] • Target: Circle theorems: tangent-chord and secant angles

Two similar solids have a volume ratio of $125:1$. Find the linear (side-length) scale factor.

a. 1:9

b. 1:10

c. **1:5** ✓

d. 1:7

Linear ratio $=\sqrt[3]{125:1}=1:5$, since $1^3:5^3=125:1$.

35 [Intermediate] • Target: Power of a point (secant and tangent segment relationships)

Three distinct vertices of a regular 12-gon are selected uniformly at random. What is the probability that the triangle formed by these three vertices is obtuse?

a. $\frac{3}{11}$

b. $\frac{5}{11}$

c. $\frac{6}{11}$ ✓

d. $\frac{7}{11}$

To find the probability that a randomly chosen triangle is obtuse, we determine the total number of possible triangles and the number of obtuse triangles. **Step 1: Total number of triangles** Choosing any 3 distinct vertices from the 12 vertices of the regular 12-gon gives: $\binom{12}{3}=\frac{12\times11\times10}{3\times2\times1}=220$ **Step 2: Counting right and acute triangles** - **Right Triangles:** By Thales' Theorem, an inscribed triangle is a right triangle if and only if one of its sides is a.

36 [Intermediate] • Target: Area of regular polygons

In ΔABC, the side lengths opposite angles $A, B,$ and C are $a, b,$ and c, respectively. Given that $a=7$, $b=8$, and $\sin(B-A)=\frac{3}{5}\sin C$, what is the length of side c?

a. 5 ✓

b. $3\sqrt{5}$

c. $\frac{5}{3}$

d. $2\sqrt{7}$

We synthesize angle sum/difference identities with the Law of Sines and the Law of Cosines. **Step 1: Relate $\sin C$ to $A+B$** Since the interior angles of a triangle sum to 180°, $C=180^\circ-(A+B)$. Therefore: $\sin C=\sin(180^\circ-(A+B))=\sin(A+B)$ Substituting this into the given equation gives: $\sin(B-A)=\frac{3}{5}\sin(A+B)$ **Step 2: Expand using compound angle formulas** Using the sine addition and subtraction identities.

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37 [Intermediate] • Target: Similar solids and scale factor in 3D

A circle has a radius of 6 units. A chord within this circle subtends a sector with a central angle of 60 degrees. What is the length of the chord?

a. 6 units ✓

b. $6\sqrt{3}$ units

c. 12 units

d. $12\sqrt{3}$ units

We have a circle with radius $r = 6$ and a sector with a central angle of 60 degrees. The chord connecting the endpoints of the arc of this sector forms an isosceles triangle with the two radii. Since the central angle is 60 degrees and the other two angles in the triangle must be equal (because the sides opposite them are radii), each of the other angles is $(180^\circ - 60^\circ)/2 = 60^\circ$. Therefore, the triangle is equilateral. In an equilateral triangle, all sides are equal. Since two sides are radii of.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Composite 3D solids • Cross-Lesson: 3.01 Advanced Functions

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?

a. $\cos\theta$ ✓

b. $\sin\theta$

c. $\tan\theta$

d. 1

By definition, the coordinates of a point $P(x, y)$ on the unit circle corresponding to an angle θ (measured counterclockwise from the positive x-axis) are given by $(\cos\theta, \sin\theta)$. Therefore, the x-coordinate of the point P is $\cos\theta$.

39 [Challenging] • Target: Triangle centers: centroid and circumcenter • Cross-Lesson: 4.01 Advanced Trigonometry

The sum of interior angles of a hexagon is:

a. 540°

b. 720° ✓

c. 900°

d. 360°

Applying Lines, Triangles & Polygons principles, $(6 - 2) \times 180 = 720°$ Therefore, the verified correct option is 720°.

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40 [Challenging] • Target: Heron's formula • Cross-Lesson: 5.01 Pre-Calculus Concepts

A circle has equation $(x-1)^2+(y+3)^2=9$. What are its center and radius?

a. Center $(-1, 3)$, radius 9

b. Center $(-1, 3)$, radius 3

c. Center $(1, -3)$, radius 9

d. **Center** $(1, -3)$**, radius** 3 ✓

Standard form $(x-h)^2+(y-k)^2=r^2$ gives center $(1, -3)$ and radius $r=3$.

41 [Challenging] • Target: A multi-part circle geometry problem • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Parallel lines have slopes that are:

a. Negative reciprocals

b. **Equal** ✓

c. Opposite signs

d. Unrelated

Parallel lines have equal slopes.

42 [Challenging] • Target: Angle bisectors and the angle bisector theorem • Cross-Lesson: 8.01 Statistics, Counting & Probability

In the diagram below, line AB is parallel to line CD. Transversal line EF intersects AB at point G and CD at point H. If the measure of angle AGH is 50°, what is the measure of angle GHD?

a. 50°

b. **130°** ✓

c. 40°

d. 90°

When a transversal intersects two parallel lines, consecutive interior angles are supplementary, meaning they add up to 180 degrees. Angle AGH and angle GHC are consecutive interior angles. Therefore, $m\angle AGH + m\angle GHC = 180^\circ$. Since $m\angle AGH = 50^\circ$, we have $50^\circ + m\angle GHC = 180^\circ$. Solving for $m\angle GHC$ gives $m\angle GHC = 180^\circ - 50^\circ = 130^\circ$. Angle GHD and angle GHC are angles on a straight line, forming a linear pair. Thus.

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43 [Challenging] • Target: The 30-60-90 and 45-45-90 triangles in composite figures • Cross-Lesson: 1.01 Advanced Number Theory

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$ ✓

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

The problem states that point $P(x, y)$ lies on the unit circle, meaning $x^2 + y^2 = 1$. We are given that the y-coordinate of P is $y = \frac{3}{5}$. Substituting this into the unit circle equation, we get $x^2 + (\frac{3}{5})^2 = 1$. This simplifies to $x^2 + \frac{9}{25} = 1$. Subtracting $\frac{9}{25}$ from both sides gives $x^2 = 1 - \frac{9}{25} = \frac{16}{25}$. Taking the square root of both sides, we get $x = \pm\frac{4}{5}$. The problem also states that point P is in the second quadrant, where the x.

44 [Challenging] • Target: Circumscribed and inscribed circles • Cross-Lesson: 2.01 Advanced Algebra & Systems

Consider a circle with its center at the origin $(0, 0)$ and a radius of 5 units. A chord of this circle is perpendicular to the x-axis and intersects the x-axis at the point $(3, 0)$. The endpoints of this chord on the circle are A and B. Determine the length of the chord AB.

a. 6

b. **8** ✓

c. 10

d. 4

The equation of the circle centered at the origin with radius 5 is $x^2 + y^2 = 5^2 = 25$. The chord is perpendicular to the x-axis and intersects it at $(3, 0)$. This means the x-coordinate of the endpoints of the chord, A and B, is 3. Substitute $x = 3$ into the circle's equation to find the corresponding y-coordinates: $3^2 + y^2 = 25$ $9 + y^2 = 25$ $y^2 = 25 - 9$ $y^2 = 16$ $y = \pm 4$ So the endpoints of the chord are $A = (3, 4)$ and $B = (3, -4)$. The length of the chord AB is the distance between these t.

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45 [Challenging] • Target: Regular polygon interior and exterior angles • Cross-Lesson: 3.01 Advanced Functions

In the diagram below, triangle ABC is a right-angled triangle with the right angle at C. CD is the altitude to the hypotenuse AB. If AD = 4 units and DB = 9 units, find the length of the altitude CD.

a. 6 units ✓

b. 13 units

c. 36 units

d. 7.5 units

In a right-angled triangle, the altitude drawn to the hypotenuse creates three similar triangles: the original triangle and two smaller triangles. Specifically, triangle ADC is similar to triangle CDB. A property derived from this similarity is that the square of the length of the altitude to the hypotenuse is equal to the product of the lengths of the two segments it divides the hypotenuse into. This is known as the Geometric Mean Theorem. Therefore, $CD^2 = AD \times DB$. Given $AD = 4$ and $DB = 9$, we have.

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